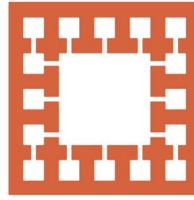


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Varendra University

Dept. of Computer Science & Engineering (CSE)

The Processor Status and the Flag Register (Chapter-5)

Course Title: Microprocessor and Assembly Language

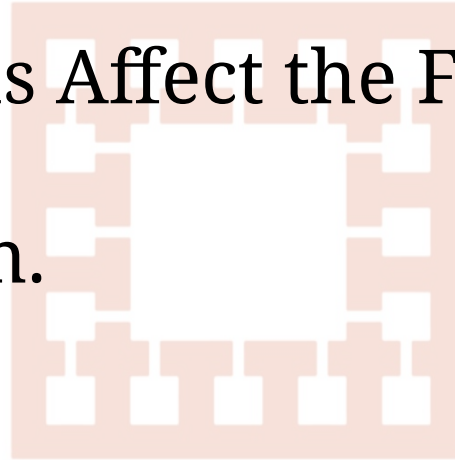
Course No: CSE 3205

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Outline

1. Flag Register.
2. Overflow.
3. How Instructions Affect the Flags.
4. Overflow Math.
5. Practice Problem.

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Flag Register

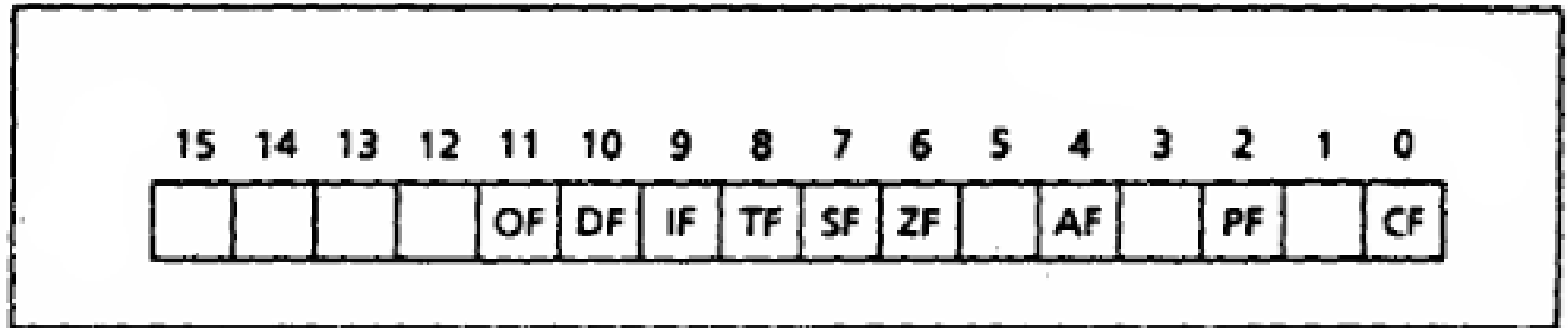
- For the 8086 processor, the processor state is implemented as nine individual bits called flags.
- Each decision made by the 8086 is based on the values of these flags.
- The flags are placed in the FLAG register and they are classified as either status flags or control flags.
- The status flags reflect the result of a computation.
- The control flags are used to enable or disable certain operations of the processor.

Flag Register

Flag register are two types,

1. Status Flag.
2. Control Flag.

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- The status flags are located in bits 0, 2, 4, 6, 7, and 11.
- The control flags are located in bit 8, 9, and 10.

Flag Register

Table 5.1 Flag Names and Symbols

Status Flags

Bit	Name	Symbol
0	Carry flag	CF
2	Parity flag	PF
4	Auxiliary carry flag	AF
6	Zero flag	ZF
7	Sign flag	SF
11	Overflow flag	OF

Control Flags

Bit	Name	Symbol
8	Trap flag	TF
9	Interrupt flag	IF
10	Direction flag	DF

Flag Register

Status Flag:

The processor uses status flags to reflect the result of an operation.

Carry Flag (CF):

- CF = 1, if there is a carry out from the most significant bit (MSB) on addition, or there is a borrow into the MSB on subtraction. Otherwise, it is 0.
- CF is also affected by shift and rotate instructions.

Flag Register

Parity Flag (PF):

- PF = 1, if the low byte of a result has an **even number of one** bits (even parity).
- It is 0 if the low byte has odd parity.

For example,

If the result of a word addition is FFFEh (**1111 1111** **1111 1110**), then the low byte contains 7 one bits, so PF = 0.

Auxiliary Flag (AF):

- AF = 1, if there is a carry out from bit 3 on addition, or a borrow into bit 3 on subtraction.
- AF is used in binary coded decimal (BCD)

Flag Register

Zero Flag (ZF):

- $ZF = 1$ for a zero result.
- $ZF = 0$ for a nonzero result.

Sign Flag (SF):

- $SF = 1$, if the MSB of a result is 1 (It means the result is negative)
- $SF = 0$, if the MSB is 0.

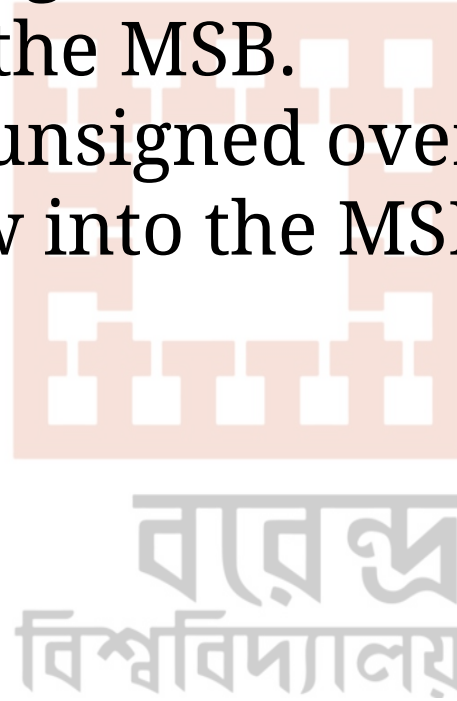
Overflow Flag (OF):

- $OF = 1$, if signed overflow occurred, otherwise it is 0.

Overflow

Unsigned Overflow:

- On addition, unsigned overflow occurs when there is a carry out of the MSB.
- On subtraction, unsigned overflow occurs when there is a borrow into the MSB.



Overflow

Signed Overflow:

- On addition of numbers with the same sign, signed overflow occurs when the sum has a different sign. **For example**, when we add 7FFFh and 7FFFh (two positive numbers), but got FFFEh (a negative result).
- Subtraction of numbers with different signs is like adding numbers of the same sign. For example, $A - (-B) = A + B$ and $-A - (+B) = -A + -B$. Signed overflow occurs if the result has a different sign than expected.
- In addition of numbers with different signs,

How Instructions Affect the Flags

<i>Instruction</i>	<i>Affects flags</i>
MOV/XCHG	none
ADD/SUB	all
INC/DEC	all except CF
NEG	all (CF = 1 unless result is 0, OF = 1 if word operand is 8000h, or byte operand is 80h)

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Overflow Math



Practice Problem

Example 5.1:

Example 5.1 ADD AX,BX, where AX contains FFFFh, BX contains FFFFh.

Solution:

$$\begin{array}{r} \text{FFFFh} \\ + \text{FFFFh} \\ \hline 1 \text{ FFFEh} \end{array}$$

The result stored in AX is FFFEh = 1111 1111 1111 1110.

SF = 1 because the msb is 1.

PF = 0 because there are 7 (odd number) of 1 bits in the low byte of the result.

ZF = 0 because the result is nonzero.

CF = 1 because there is a carry out of the msb on addition.

OF = 0 because the sign of the stored result is the same as that of the numbers being added (as a binary addition, there is a carry into the msb and also a carry out).

Practice Problem

Example 5.2:

Example 5.2 ADD AL,BL, where AL contains 80h, BL contains 80h.

Solution:

$$\begin{array}{r} 80h \\ + 80h \\ \hline 100h \end{array}$$

The result stored in AL is 00h.

SF = 0 because the msb is 0.

PF = 1 because all the bits in the result are 0.

ZF = 1 because the result is 0.

CF = 1 because there is a carry out of the msb on addition.

OF = 1 because the numbers being added are both negative, but the result is 0 (as a binary addition, there is no carry into the msb but there is a carry out).

Practice Problem

Example 5.3:

Example 5.3 SUB AX,BX, where AX contains 8000h and BX contains 0001h.

Solution:

$$\begin{array}{r} 8000h \\ - 0001h \\ \hline 7FFFh = 0111\ 1111\ 1111\ 1111 \end{array}$$

The result stored in AX is 7FFFh.

SF = 0 because the msb is 0.

PF = 1 because there are 8 (even number) one bits in the low byte of the result.

ZF = 0 because the result is nonzero.

CF = 0 because a smaller unsigned number is being subtracted from a larger one.

Now for OF. In a signed sense, we are subtracting a positive number from a negative one, which is like adding two negatives. Because the result is positive (the wrong sign), OF = 1.

Practice Problem

Example 5.4:

Example 5.4 INC AL, where AL contains FFh.

Solution:

$$\begin{array}{r} \text{FFh} \\ + \text{1h} \\ \hline \text{1 00h} \end{array}$$

The result stored in AL is 00h. SF = 0, PF = 1, ZF = 1. Even though there is a carry out, CF is unaffected by INC. This means that if CF = 0 before the execution of the instruction, CF will still be 0 afterward.

OF = 0 because numbers of unlike sign are being added (there is a carry into the msb and also a carry out).

Practice Problem

Example 5.5:

Example 5.5 `MOV AX, -5`

Solution: The result stored in AX is `-5 = FFFBh`.

None of the flags are affected by MOV.



Practice Problem

Example 5.6:

Example 5.6 NEG AX, where AX contains 8000h.

Solution:

$$\begin{array}{r} 8000\text{h} = 1000\ 0000\ 0000\ 0000 \\ \text{one's complement} = 0111\ 1111\ 1111\ 1111 \\ \qquad \qquad \qquad + 1 \\ \hline 1000\ 0000\ 0000\ 0000 = 8000\text{h} \end{array}$$

The result stored in AX is 8000h.

SF = 1, PF = 1, ZF = 0.

CF = 1, because for NEG CF is always 1 unless the result is 0.

OF = 1, because the result is 8000h; when a number is negated, we would expect a sign change, but because 8000h is its own two's complement, there is no sign change.

In the next section, we introduce a program that lets us see the actual setting of the flags.